Application of an adapted Verlet algorithm to the numerical analysis of chaotic dynamics in the Duffing oscillator

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For decades, non-linear systems, such as the Duffing oscillator, were difficult to solve or fully understand, despite their chaotic behaviour being all around us; from weather forecasting to the three body-problem. This research investigated and quantified the dynamical behaviour of the Duffing oscillator for specific parameter values using an adapted version of the Verlet algorithm with the aim of gaining insights into chaotic phenomena in non-linear system as well as to examine the computational accuracy of the Verlet algorithm. Using this algorithm, various parameters in the Duffing equation were investigated by analysing the standard deviation of the error using a conservative Hamiltonian, and its chaos was quantified by calculating the fractal dimension using a box-counting method and linear regression in Poincaré sections. The results indicated that the Verlet algorithm introduced the presence of shadow Hamiltonians, and revealed that certain values for parameters in the Duffing equation can effect the accuracy. Additionally, the fractal dimension yielded a value of 1.782 ± 0.002, along with an R^2 value of 0.998. Overall, this study showed that the Verlet algorithm can be used to investigate the behaviour of the Duffing oscillator while also highlighting the analytical limitations.

I. INTRODUCTION

Until recent decades, non-linear systems were treated as rare oddities and were mostly avoided through linear approximations, not only because they were difficult to solve and understand, but also due to the computational roadblocks that made it impossible to analyse these systems and apply them in real-life practices [1]. However, non-linearity is fundamental and describes nearly every physical system [2]. Therefore by reducing numerical analysis to linear approximations, only a small portion of the whole picture is covered, since linear and non-linear systems do not share the same characteristics, such as the superposition principle [1][2]. Sometimes even the smallest deviations from the initial conditions can cause major diverging outcomes for varying dynamical systems, making predictions in the far future impossible [3]. This sensitivity to initial conditions in non-linear systems is one of the many characteristics of chaos along with its erratic and non-periodic long-term behaviour, showing that a deterministic system can still exhibit an unpredictable future [1]. Although this phenomenon became more manageable to study with the emergence of computers and the rise of their computational power, it remains to this day impossible to solve many related problems in practise, ranging from astronomical scales such as the three-body problem to applications like in weather forecasting [3].

A widely known and used non-linear system that can exhibit this type of chaotic behaviour is the Duffing oscillator, as the dynamics of such systems are comparable to those observed in the natural world [3]. Despite its chaotic behaviour, the system eventually settles into an equilibrium pattern over time, known as the strange attractor, showing that even in complex systems, order can still emerge [4].

In this research, an adapted version of the Verlet algorithm is used to analyse the dynamics of the Duffing oscillator, with the aim of quantifying chaotic behaviour for specific parameter values. This is achieved by examining the shapes of the strange attractors, also known as fractals, in order to gain insight into chaotic phenomena in non-linear systems such as the Duffing oscillator, as well as to examine computational accuracy of the Verlet algorithm [5].

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II. THEORY

A. Duffing Oscillator

The Duffing oscillator is essentially a non-linear oscillator model that includes a cubic stiffness term in the differential equation describing an ideal linear harmonic oscillator as depicted in Equation 1 [4][6].

$$F[x, \dot{x}, t] = -\gamma v(t) + 2ax(t) - 4bx^3(t) + F_0 \cos(\omega t) \quad (1)$$

with the first term including γ which contributes to the velocity-dependent friction, the second and third term containing a and b representing the force the particle feels when it moves in a double potential well, and the last term which involves the external periodic force along with the driving frequency ω and the oscillation amplitude F_0. For particular values for the given parameters, the Duffing oscillator will reveal chaotic behaviour, meaning erratic and non periodic long-term behaviour, where small changes in its initial behaviour cause varying outcomes [3].

B. Fractal Dimension

Chaotic behaviour in non-linear oscillators can be visualized using Poincaré sections, which essentially represent the motion of a particle sampled at intervals of one cycle [1]. Poincaré sections of chaotic motion reveal patterns that are intrinsic to the system and are known as so-called fractals. Fractals are geometric structures with fractional dimensions, whose values are often used to quantify spatial complexity [7][8]. These structures exhibit self-similarity; when zooming in, similar patterns reappear on smaller scales. This process can in principle continue forever, as observed in nature, e.g. snowflakes and broccoli, but as well as in mathematical structures such as the famous Mandelbrot fractal. Whenever such characteristic is observed in Poincaré sections, the long term motion of the particle is said to be a strange attractor [1].

C. Verlet Algorithm

The Verlet algorithm is a method used for integrating second-order differential equations of the form in Equation 2

$$m\ddot{x}(t) = F[x(t), t] \tag{2}$$

where for the rest of this paper, the mass (m) is set to 1. The method includes evaluating the position x together with the force F at discrete time interval steps. The algorithm can be derived by using Taylor approximations for x at $t = t \pm h$.

$$x(t+h) = x(t) + hv(t) + \frac{h^2}{2}F[x(t), t] + \frac{h^3}{6}\dddot{x}(t) + O(h^4) \tag{3}$$

$$x(t-h) = x(t) - hv(t) + \frac{h^2}{2}F[x(t), t] - \frac{h^3}{6}\dddot{x}(t) + O(h^4) \tag{4}$$

Adding Equations 3 and 4 together, the algorithm can be obtained as shown in Equation 5.

$$x(t+h) = 2x(t) - x(t-h) + h^2F[x(t), t] + O(h^4) \tag{5}$$

However, when only the initial conditions for the position and velocity at a given time t are known, the algorithm needs to be rewritten in terms of parameters with known values at positive time steps. This is done by subtracting Equations 3 and 4 in order to derive the velocity at integer time steps, as shown in 6.

$$v(t) = \frac{x(t+h) - x(t-h)}{2h} + O(h^2) \tag{6}$$

Subsequently, evaluating the velocity at half integer steps instead and dropping the error term for now, Equation 7 is obtained.

$$v(t - \frac{1}{2}h) = \frac{x(t) - x(t-h)}{2(\frac{1}{2}h)} \tag{7}$$

As a result, Equation 5 can now be expressed as shown in Equation 8.

$$\begin{aligned} x(t+h) &= x(t) + h\frac{x(t) - x(t-h)}{h} + h^2F[x(t), t] \\ &= x(t) + hv(t - \frac{1}{2}h) + h^2F[x(t), t] \end{aligned} \tag{8}$$

Consequently, the Verlet algorithm can be reformulated into a so-called leap-frog form as represented in Equation 9 and 10.

$$v(t + h/2) = v(t - \frac{1}{2}h) + hF[x(t), t] \tag{9}$$

$$x(t+h) = x(t) + hv(t + \frac{1}{2}h) \tag{10}$$

Lastly, to get rid of the negative time step, Taylor expansion is used one last time for v at $t = t - \frac{1}{2}h$ as shown in Equation 11.

$$v(t - \frac{1}{2}h) = v(t) - \frac{1}{2}hF[x(t), t] + O(h^2) \tag{11}$$

Additionally, doing the same to determine the velocity at integer steps, the final form of the Verlet algorithm can be derived as shown in Equations 12, 13 and 14.

$$v(t + h/2) = v(t) + \frac{1}{2}hF[x(t), t] \tag{12}$$

$$x(t+h) = x(t) + hv(t + \frac{1}{2}h) \tag{13}$$

$$v(t+h) = v(t + h/2) + \frac{1}{2}hF[x(t+h), t+h] \tag{14}$$

1. Error Accummulation

To analyse the accumulated error, the Taylor expansion around $t = t + h$ is again considered. Due to the time invariance of the algorithm, Equation 6 can be used in this expression as shown in Equation 15.

$$\begin{aligned} x(t+h) &= x(t) + hv(t) + \frac{h^2}{2}F[x(t), t] + O(h^3) \\ &= x(t) + h(\frac{x(t+h) - x(t-h)}{2h} + O(h^2)) + \frac{h^2}{2}F[x(t), t] \\ &= x(t) + \frac{1}{2}x(t+h) - \frac{1}{2}x(t-h) + \frac{h^2}{2}F[x(t), t] + O(h^2) \\ &= 2x(t) - x(t-h) + h^2F[x(t), t] + O(h^2) \end{aligned} \tag{15}$$

As a result, the equation is in the same form as Equation 5. Therefore, the error in the position of the leap-frog form is still $O(h^4)$.

2. Stability

In order to analyse the stability, the analytical solution of the algorithm is derived using the Verlet solution for a harmonic oscillator $\ddot{x}(t) = -Cx(t)$, along with the general analytical solution $x(t) = e^{i\omega t}$, as derived in Equation 17.

$$\begin{aligned} &x(t+h) = 2x(t) - x(t-h) - h^2Cx(t) \\ &e^{i\omega(t+h)} = 2e^{i\omega t} - e^{i\omega(t-h)} - h^2e^{i\omega t} \\ &e^{i\omega(h)} + e^{-i\omega(h)} = 2 - h^2C \qquad (16) \\ &2 - 2\cos(\omega h) = h^2C \\ &\cos(\omega h) = 1 - \frac{1}{2}h^2C \qquad (17) \end{aligned}$$

Since the range of the cosine function is $[-1, 1]$, it follows that $0 \leq h^2C \leq 4$. This range holds for any $\omega \in \mathbb{R}$. When $h^2C > 4$, a real ω no longer exists, and the oscillatory behaviour of Equation 16 becomes exponential, involving a positive exponential term $e^{+\phi h}$, where $\omega = i\phi$, that grows to infinity. Therefore, the analytical solution becomes unstable for $h^2C > 4$. In conclusion, h needs to be sufficiently small in order to avoid having instability in actual applications.

3. *Adapted Duffing Verlet Algorithm*

In order to solve for the Duffing oscillator using the Verlet algorithm, additional steps must be realized since the Duffing equation contains a velocity term. Equation 1 is therefore separated into Equation 18.

$$F(x, \dot{x}, t) = F(x(t), t) - \gamma v(t) \tag{18}$$

By substituting this into the integer and the half integer velocity Verlet algorithm Equations 12 and 14, the following Equations 19 and 20 can be derived.

$$\begin{aligned} v(t + \frac{1}{2}h) &= v(t) + \frac{1}{2}h(F(x, t) - \gamma v(t)) \\ &= (1 - \frac{1}{2}\gamma h)v(t) + \frac{1}{2}hF(x(t), t) \end{aligned} \tag{19}$$

$$\begin{aligned} v(t + h) &= v(t + h/2) + \frac{1}{2}h(F(x(t + h), t + h) - \gamma v(t + h)) \\ (1 + \frac{1}{2}h\gamma)v(t + h) &= v(t + \frac{1}{2}h) + \frac{1}{2}F(x(t + h), t + h) \\ v(t + h) &= \frac{v(t + \frac{1}{2}h) + \frac{1}{2}F(x(t + h), t + h)}{1 + \frac{1}{2}\gamma} \end{aligned} \tag{20}$$

These formulations, together with the position in Equation 13, form the adapted Verlet algorithm used to numerically solve the Duffing oscillator.

III. METHOD

A. Conservative Hamiltonian Analysis

To verify whether the algorithm for solving the Duffing oscillator works as intended, the Hamiltonian of a conservative system was calculated using Equation 1 along with $a = 0.5$, $b = 0.25$, $t \in [0, 50]$ with an interval step size of $h = 5 \cdot 10^{-4}$, and by setting the non-conservative parameters γ and F_0 to zero. To analyse the error in the Verlet algorithm, Equation 21 was used to isolate the error.

$$\varepsilon_H = H(t) - H(t = 0) \tag{21}$$

with $H(t)$ the hamiltonian at time t, $H(0)$ the initial hamiltonian at $t = 0$, and ε_H the error. Subsequently, the error was plotted as a function of time to analyse the effects of the Verlet algorithm. A theoretical prediction for the cause of the error due to the Verlet algorithm, is the shadow Hamiltonian as given in Equation 22 as derived by Toxvaerd [9]. This is essentially a Hamiltonian that arises from discrete time calculations in the Verlet algorithm, causing it to fluctuate around the real physical Hamiltonian value.

$$\tilde{H} = H + h^2 f(x, v) + \mathcal{O}(h^4) \tag{22}$$

with $\tilde{H}$ the shadow hamiltonian, H the analytical Hamiltonian, h the integration time step, and $f(x, v)$ an arbitrary function. This definition leads to an error as described in Equation 23.

$$\tilde{\varepsilon}_H = h^2 f(x, v) + \mathcal{O}(h^4) \tag{23}$$

with $\tilde{\varepsilon}_H$ the error of the shadow Hamiltonian. This was analysed by calculating the standard deviation for different h values and plotting them on a log-scale to obtain the slope using linear regression, which is expected to be 2 according to Equation 23. The values for h used were chosen between $[10^{-6}, 10^{-2}]$ with logarithmic spacing and the corresponding goodness of fit R^2 value was determined.

Furthermore, the shape of the error ε_H was analysed by calculating and plotting the standard deviations for different values of the Duffing parameters a and b. The values used for a and b were logarithmically spaced in $[10^{-3}, 10^1]$ and $[10^{-4}, 10^3]$, respectively.

B. Sensitivity To Initial Conditions Analysis

The Duffing oscillator equation was solved using the adapted Verlet algorithm as described in Equations 13, 19 and 20 using the following parameters: $a = 0.5$, $b = 0.25$, $F_0 = 2$, $\gamma = 0.1$, $\omega = 2.4$, $t \in [0, 50]$ with an interval step size of $h = 5 \cdot 10^{-4}$. The initial conditions were first taken as $v_0 = 0$ with five values for x_0 ranging between $[0.499, 0.501]$ in equal steps. Subsequently, the time-position and the phase domain were plotted and analysed. Similar analysis was conducted by varying five initial velocity v_0 values between $[-0.01, 0.01]$ in equal steps, and keeping x_0 constant at 0.5.

C. Strange Attractor Analysis

In order to analyse the strange attractor of the Duffing oscillator, the Verlet algorithm was once again used, along with $x_0 = 0.5$, $v_0 = 0$, $a = 0.5$, $b = 0.25$, $F_0 = 2$, $\gamma = 0.1$, $\omega = 2.4$, with time $t \in [0, \frac{N \cdot T}{N_T}]$ and a step size h of $\frac{T}{N_T}$, where T is the period calculated using ω, $N = 25 \cdot 10^6$, and $N_T = 10^2$. By doing this, the x and v values can be extracted and plotted in a Poincaré section for each time a period T has elapsed.

Furthermore, the fractal dimension of the resulting attractor was analysed using a box-counting method. First, the shape of the attractor was normalized by shifting and scaling it into a square of side length 1. Next, the square was divided into smaller $l \times l$ square cells, where $l = 2^n$ with $\{n \in \mathbb{Z} \mid 1 \leq n \leq 8\}$. A higher maximum value for in the range of l will introduce saturation effects due to the fact that only a finite amount of particles were plotted. To continue, each point was assigned to a specific cell by multiplying the normalized attractor points by l and flooring the result to an integer value, resulting in coordinates for the cell squares. By filtering out duplicate entries, only the number of squares containing at least one attractor point was obtained. The relationship used in the box-counting method is expected to follow Equation 24.

$$N(c) \propto c^{-D_f} \tag{24}$$

where D_f is the dimension, c the side of the cells $1/l$, and $N(c)$ the number of squares. Subsequently, linear regression was used to verify the relation with the calculated data, and the corresponding R^2 value were determined. This method can be intuitively understood by examining how the number of covering boxes scale with their sizes. For a one-dimensional line segment of length r, covering it with boxes of size c gives $N(c) = \frac{r}{c}$. Scaling the box size by a factor s results in the number of boxes scaling inversely, corresponding to $D_f = 1$. Similarly, for a two-dimensional square of side length r, $N(c) = \frac{r^2}{c^2}$, and scaling yields an inversely quadratic relation, consistent with $D_f = 2$. In general, for a self-similar object, the number of boxes therefore follow Equation 24.

IV. RESULTS AND DISCUSSION

A. Conservative Hamiltonian Analysis Results

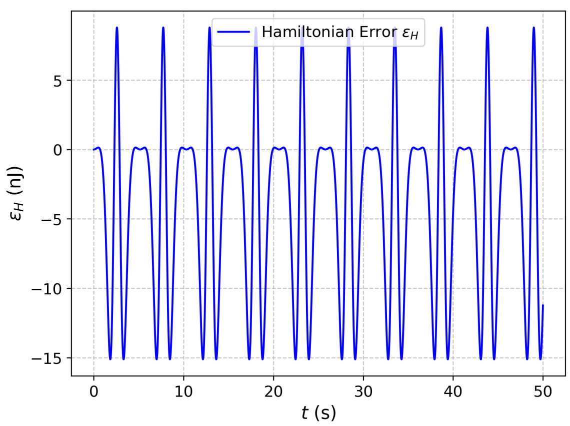

FIG. 1. A plot showing the time t versus the Hamiltonian error ε_H of the Verlet algorithm solving a conservative Hamiltonian. The error was calculated by taking the difference between the intial Hamiltonian and the Hamiltonian at a given time t.

Figure 1 shows the error in the Hamiltonian over time as a result of the analysis of the Verlet algorithm on a conservative system. Considering that the total Hamiltonian was calculated to be around -0.1 J, the observed error with peak to peak amplitude of 20 nJ is negligible. The shape of the error over time shows distinct periodical behaviour, resonating with the hypothesis of a shadow hamiltonian as discussed in Section III A.

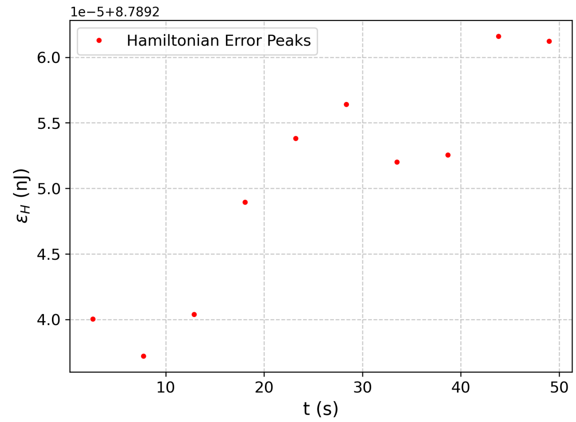

FIG. 2. A plot showing the time t versus the Hamiltonian error ε_H of the Verlet algorithm solving a conservative Hamiltonian. The red dots represent isolated top peak error values, which are not constant in time, indicating a change of the Hamiltonian error over time.

Figure 2 shows a closer examination of the observed error of the top peaks as a function of time, revealing an error growth. However, the error difference between the first peak and the last peak is $2.1 \cdot 10^{-14}$ nJ, and thus negligible relative to the total Hamiltonian.

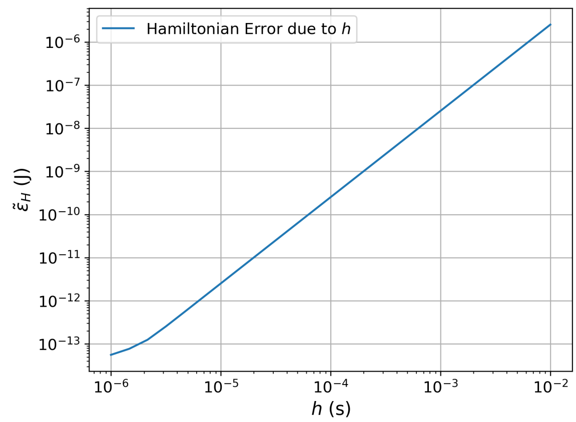

FIG. 3. A logarithmic plot showing the interval time step size h versus the shadow Hamiltonian error $\tilde{\varepsilon}_H$ of the Verlet algorithm solving a conservative Hamiltonian. A linear regression analysis yielded a slope of 1.96 ± 0.02 along with an R^2 value of 0.999.

Figure 3 shows the resulting standard deviations for different interval time steps h, yielding a slope of 1.96 ± 0.02, which is close to the expected value of 2, along with a R^2 value of 0.999. This verifies the hypothesis involving the occurrence of shadow Hamiltonians as discussed in Section III A.

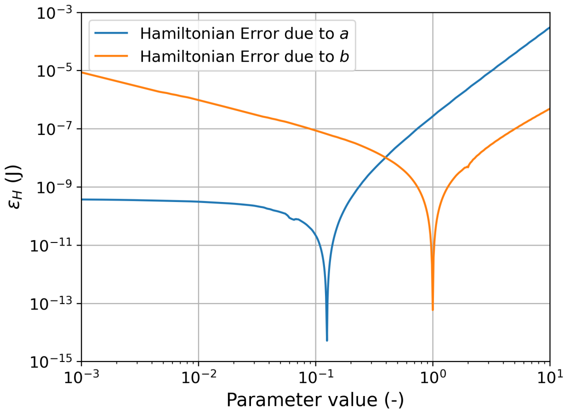

FIG. 4. A logarithmic plot showing the conservative Hamiltonian error ε_H of the Verlet algorithm versus both the Hamiltonian position (x) constant a (blue line), and the Hamiltonian cubic position term (x^3) constant b (orange line). The minimum error values occur at $a = 0.125$ and $b = 1.002$.

Additionally, it is seen in Figure 4, that the error yields minima at a=0.125, and at b=1.002. This is because there are no oscillations for these values as the initial position is at an equilibrium point. It can be seen that while the error grows as b becomes smaller past the equilibrium point, a reaches a constant value to the left of the equilibrium point, which was approximated to be $\varepsilon_H = 3.7 \cdot 10^{-10}$. This is because as a goes to zero, the potential becomes a simple attractive potential at which the particle will oscillate. When b goes to zero the potential becomes a repulsive potential and the potential becomes unstable causing high particle speeds, resulting in a growth in error again.

B. Sensitivity To Initial Conditions Analysis Results

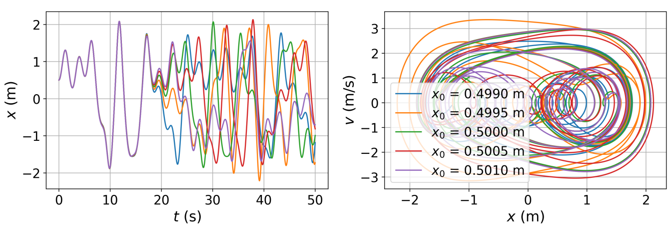

FIG. 5. The figure shows plots of the solutions to the Duffing oscillator using an adapted Verlet algorithm for different initial conditions x_0. The left plot shows the spatial domain (t, x) and the right shows the phase space (x, t).

Figure 5 shows the results of the simulations for slightly differing initial conditions in x_0 for the Duffing oscillator. It can be seen that, even though the solutions initially follow the same path, the solutions become completely uncorrelated. This confirms the high sensitivity to initial conditions in the Duffing equation.

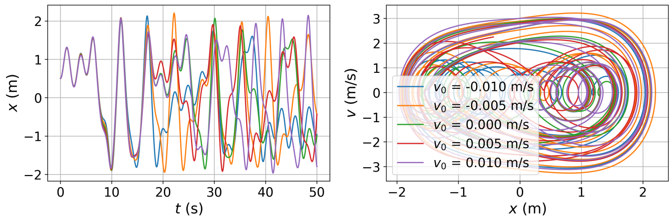

FIG. 6. The figure shows plots of the solutions to the Duffing oscillator using an adapted Verlet algorithm for different initial conditions v_0. The left plot shows the spatial domain (t, x) and the right shows the phase space (x, t).

Figure 6 shows similar simulations for small changes in the initial velocity v_0, and it again shows chaos setting in at later times. The interval time step sizes where the chaos became more noticeable was $h > 1$.

C. Strange Attractor Analysis Results

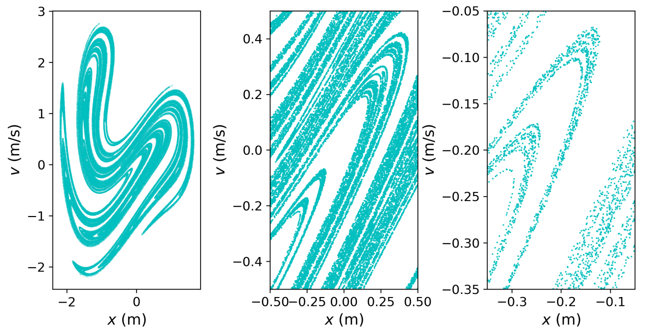

FIG. 7. The figure shows Poincaré sections of the solutions to the Duffing oscillator using an adapted Verlet algorithm. The first plot shows an overall picture of the Duffing attractor, and the second and third plots show zoomed in sections of the attractor.

Figure 7 shows the Poincaré sections of the solutions obtained at each time a period T had elapsed. The result reveals a distinct attractor, similar to those found in literature [10]. At first glance, the attractor appears to exhibit ordered, non-chaotic behaviour. However, upon closer inspection, chaotic structure emerges, giving it an increasingly random pattern.

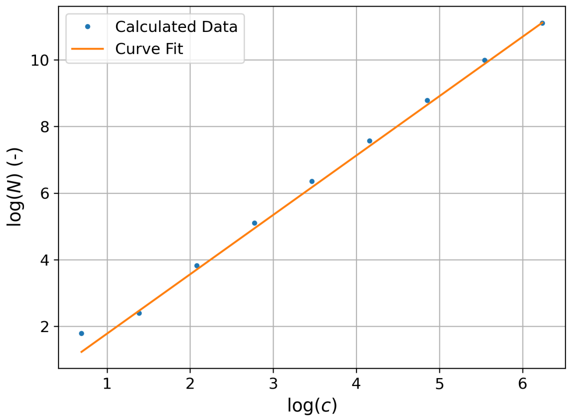

FIG. 8. The plot shows results from a box-counting analysis of a Duffing oscillator used to determine its fractal dimension. The y-axis represents the logarithm of the number of $l \times l$ squares covering the attractor, while the x-axis shows the logarithm of the scaling factor $c = 1/l$. The blue dots indicate the extracted data points, and the orange line shows the linear regression fit, yielding a fractal dimension of $D_f = 1.782 \pm 0.002$ with an R^2 value of 0.998.

Figure 8 shows the results of the fractal dimension analysis of the Duffing attractor, yielding $D_f = 1.782 \pm 0.002$, along with an R^2 value of 0.998. Since the obtained dimension is fractal, the shape can be characterized as a strange attractor.

D. Recommendations

Although, the analysis of the conservative Hamiltonian using the Verlet algorithm showed that the error due to shadow Hamiltonians proved to be negligible with the interval time steps h used, it must be noted that the Duffing oscillator is a non-conservative system with non-conservative terms that were not considered during the analysis. Future studies could potentially examine more closely the deviations introduced by these dissipative terms. Additionally, the conservative Hamiltonian isolated error peaks appear to exhibit oscillatory behaviour with an increasing central equilibrium value. However, to confirm this, the behaviour should be analysed over a longer time scale to enable proper model fitting and further investigation of its underlying cause.

Another point to note is that according to literature, values for the Duffing equation parameters influence the fractal dimension results; higher damping seems to reduce the fractal dimension of non-linear systems [11][12]. Not only could this be further investigated, but the accuracy of the Verlet method could also additionally be quantified by studying the bifurcation diagrams and determining the Feigenbaum number [1][11].

V. CONCLUSION

This study used an adapted version of the Verlet algorithm to analyse the dynamics of the Duffing oscillator. The goal was to evaluate the effectivity of the algorithm, along with the aim to quantify the chaotic behaviour of the Duffing equation. This was achieved by investigating the influence of different parameters values, analysing the standard deviation of the error using a conservative Hamiltonian, and by investigating the fractal dimension using a box-counting method. The results showed a presence of shadow hamiltonians, and revealed that certain parameters in the Duffing equation can effect the accuracy of the Verlet algorithm. Furthermore, the fractal dimension of the Duffing attractor yielded a value of 1.782 ± 0.002, along with an R^2 value of 0.998, indicating it can be characterized as a strange attractor. Future studies could examine the influence of the dissipative parameters in more detail and further investigate the time evolution of the error, particularly its possible oscillatory behaviour with an increasing equilibrium value. Additionally, the effect of different damping values on the fractal dimension could be analysed, alongside an assessment of the Verlet algorithm's accuracy through bifurcation diagrams of the Duffing solutions. Overall this study indicated that the Verlet algorithm is a useful method for investigating the behaviour of non-linear systems such as the Duffing oscillator, while also highlighting its analytical limitations.

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